JAIN DEEMED-TO-BE UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY

Teaching Learning Evaluation Plan (TLEP)

Academic Year: 2026-2027 (ODD)

Name of the Department/ School	Computer Science & Engineering
Programme	Computer Science & Engineering (AIML)
Year / Semester	II / Third
Course Code and Title	25ESHS01 - Design Thinking
Credit Value	2
Total Notional Hours Available	60 Hrs
LTPE Distribution	L: T: P: E:: 0:0:4:0
Assessment Model	100% Continuous Assessment (80:20)
Number of CA Components	2 (Unit-wise Exercise Portfolio; Comprehensive Design Challenge Project)
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Faculty In-charge	Dr. Sujeet S. Jagtap

Faculty In-Charge **Course Coordinator** **Programme Head**

Program Outcomes

PO#	Statements
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems that integrate socio-economic implications of technical applications.
PO 2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development (WK1 to WK4), applying epistemic skills that connect theoretical frameworks with real-world experiences.
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required (WK5).
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using critical inquiry methodologies and research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions (WK8), where stakeholders participate as co-creators in the knowledge development process.
PO 5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & AI based IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment (WK1, WK5, and WK7), to drive social, economic, and political transformations.
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws (WK9), integrating equality, democratic behavior, tolerance, and associated living principles.
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams, applying conjoint communicated experience and democratic engagement.
PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large exhibiting emotional intelligence, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language and learning differences
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles, innovative problem-solving approaches and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary and entrepreneurial environments.
PO 11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8), contributing to human development. Knowledge and Attitude Profile (WK)

Program Specific Outcomes (PSO)

Program	CSE – AIML	
CSE – AIML	**PSO-1**	To design and develop innovative software and hardware solutions by applying principles of Computer Science and Engineering, addressing both industry and societal needs through cutting-edge technologies.
	PSO-2	To utilize modern tools and technologies in emerging domains to solve real-world problems, with a strong emphasis on ethical considerations and societal impact.
	PSO-3	To apply AI/ML engineering concepts of data processing, model development, and intelligent decision-making to develop safe, secure, and robust production-ready AI/ML solutions.

Knowledge and Attitude Profile

WK#	Statements
WK 1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK 2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK 3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK 4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK 5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK 6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK 7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK 8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK 9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Year Wise Learning Outcomes: Second Year – Development Level

Dimension	LO #	Learning Outcome Statements
Critical Thinking	**LO1**	Challenge existing knowledge through structured analysis
	LO2	Apply systematic observation methods to engineering problems
	LO3	Evaluate technical information from multiple sources
	LO4	Develop reasoned arguments for engineering decisions and corroborate using AI
	LO5	Perceive objects and knowledge
	LO6	Raise doubts on status quo of objects and knowledge
Innovation and Professional Development	**LO1**	Execute project-based learning assignments independently
	LO2	Discuss and record observations of industrial visits
	LO3	Understand design thinking and its cultural and social implications
	LO4	Use Generative AI to visualize real-world problem
Research Experience	**LO1**	Conduct comprehensive literature reviews using AI-powered tools
	LO2	Define, identify, justify and articulate research problems using AI and develop basic research proposals
	LO3	Structure mini-projects using appropriate research methods
	LO4	Communicate effectively with industry experts and researchers related to relevant domain/ projects
Real World Exposure	**LO1**	Demonstrate interpersonal and social skills to constructively interact with society and industries
	LO2	Observe the real-world society, industry, market and insightfully infer
	LO3	Participate in group discussion and demonstrate social communication skills
	LO4	Depict the character required to practice togetherness and demonstrate team working skills
	LO5	Debate and depict tolerance on controversies and opposing ideas
	LO6	Engage meaningfully with industrial and societal partners
	LO7	Understand affective and cultural aspects of design thinking in engineering developments
	LO8	Demonstrate understanding of professional standards and professional ethics
	LO9	Show awareness of industrial practices and requirements
Youth Development	LO1	Practice social skills including displaying tendencies and attitudes towards cultural sensitivities, empathy and compassion by cultural activities, clubs and social events
	LO2	Interact effectively with general public, community leaders and NGOs
	LO3	Practice healthy food habits and healthy lifestyle habits leading to physical fitness
	LO4	Apply civic responsibilities in a given civic situation

Dimension	LO #	Learning Outcome Statements
	LO5	Practice civic and political conscience by abiding the constitution, laws and legislation, democratic behavior and ensuring social harmony
	LO6	Create awareness of social implications of engineering solutions using AI

Course Description

A. Prerequisite

No formal prerequisite is stated in the syllabus. Basic Python programming and introductory machine-learning awareness are desirable for the small language model (SLM) build track; the design-thinking work itself requires no prior technical knowledge.

B. Course Description

This foundational course introduces second-year engineering students to human-centred design thinking, with a focus on technology that serves diverse communities, particularly underserved populations. Students move through the five stages – Empathise, Define, Ideate, Prototype, Test – and then connect design outcomes to entrepreneurship and social innovation. The course supports experiential learning, ethical design values and multidisciplinary approaches, and connects with Product Innovation & Entrepreneurship (PCC 311) and Research Methods and Design (PCC 306).

For the CSE (AIML) programme the course is delivered through one continuing design journey. Weeks 1–3 build the general design-thinking foundations using the CSE design lens (digital products and human-software interaction). From Week 4 each student chooses a domain and dataset, discovers a real user problem, and re-designs one or more components of the Transformer architecture to fit that dataset. By the end of the semester each student has **built, tested and presented a small language model (SLM) trained on a dataset of their chosen domain**, together with a design journal, a model card and a business model canvas. This artefact is the starting point for the track the student takes in later semesters: **Professional, Entrepreneur or Research**.

C. Impact / Application Areas

The skills developed here apply to product design, human-centred AI, domain-specific and on-device language models, inclusive digital services, start-ups and social enterprises, and research in language modelling for under-served domains and languages.

Course Learning Objectives

The aim of this course is to:

1) Apply human-centred design principles to identify and solve real-world problems.
2) Demonstrate empathy for diverse user groups, especially underserved communities.
3) Develop solutions that consider economic, social and technical constraints and encourage low-cost, sustainable and inclusive technology solutions.
4) Work collaboratively in interdisciplinary teams.
5) Apply design thinking across the innovation cycle – empathy, ideation, prototyping and validation – to create inclusive and sustainable solutions.

Course Outcomes

Upon successful completion of this course, students will be able to:

CO #	Course Outcome statement	Bloom's Level
CO1	**Identify** the fundamental principles and stages of design thinking methodology through experiential learning exercises.	Understanding (L2)
CO2	**Demonstrate** empathy mapping and user observation techniques to capture emotional aspects and conduct basic user research, including individual interviews.	Applying (L3)
CO3	**Generate** innovative solutions through systematic ideation processes that incorporate social experience and market perspectives.	Creating (L6)
CO4	**Construct** proof-of-concept prototypes using appropriate tools and techniques for initial solution validation.	Creating (L6)
CO5	**Evaluate** design concepts using testing methodologies and assess implementation feasibility for production planning.	Evaluating (L5)
CO6	**Recognize** connections between design thinking outcomes and entrepreneurship opportunities for start-up and social innovation development.	Understanding (L2)

CO-PO/PSO Mapping

COs\POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	-	2	-	-	-	1	-	-	-	-	2	-	-	-
CO2	-	3	-	3	-	3	2	2	2	-	-	-	2	-
CO3	-	2	3	-	-	2	-	3	-	-	1	3	-	2
CO4	2	-	3	-	3	-	-	2	-	-	-	3	3	3
CO5	-	-	2	3	2	2	2	-	1	-	-	-	-	3
CO6	-	-	-	-	-	2	2	-	3	3	3	-	2	-
Average	**2.00**	**2.33**	**2.67**	**3.00**	**2.50**	**2.00**	**2.00**	**2.33**	**2.00**	**3.00**	**2.00**	**3.00**	**2.33**	**2.67**

Scale: 3 – High, 2 – Moderate, 1 – Low, – no direct mapping. The syllabus does not give a CO-PO matrix; this mapping is proposed for faculty and Board review.

PO	Mapped COs and Strength	Rationale and Assessment Evidence
PO1	CO4 (Moderate)	Applying Transformer, language-modelling and data-processing fundamentals to build the proof-of-concept SLM.
PO2	CO1 (Moderate); CO2 (High); CO3 (Moderate)	Framing a domain problem from user evidence, root-cause analysis (5 Whys) and converting insight into a point-of-view and How Might We statement.
PO3	CO3 (High); CO4 (High); CO5 (Moderate)	Generating, sketching and prototyping architectural and training-objective changes for a specific dataset and user need.
PO4	CO2 (High); CO5 (High)	Planning and conducting interviews, surveys and ablation experiments; interpreting results against user-defined success measures.
PO5	CO4 (High); CO5 (Moderate)	Using modern ML frameworks, notebooks, version control and compute resources for dataset preparation, SLM training and evaluation.
PO6	CO1 (Low); CO2 (High); CO3 (Moderate); CO5 (Moderate); CO6 (Moderate)	Understanding the social, cultural and affective context of users, including underserved communities, and the societal effect of language technology.

PO	Mapped COs and Strength	Rationale and Assessment Evidence
PO7	CO2 (Moderate); CO5 (Moderate); CO6 (Moderate)	Consent, anonymity and licensing in user research and datasets; bias, privacy and responsible AI in the model card and pitch.
PO8	CO2 (Moderate); CO3 (High); CO4 (Moderate)	Team-based empathy work, collective ideation, peer review and shared prototype development.
PO9	CO2 (Moderate); CO5 (Low); CO6 (High)	Interviewing, documenting, storyboarding and pitching to non-technical audiences.
PO10	CO6 (High)	Business model canvas, feasibility (technical, economic, social), SMART metrics and start-up planning.
PO11	CO1 (Moderate); CO3 (Low); CO6 (High)	Reflective practice, learning new methods and tools independently, and choosing a future learning track.

Pedagogical Approach and Learning Activities

CO #	Suggested Pedagogical Approach	Skill Acquired	Learning Activity (Brief Description)	Dimension Mapping	Learning Outcome Mapping
CO1	Interactive lectures, Think-Pair-Share and the Kolb experiential cycle; CSE design lens on digital products.	Conceptual understanding; comparing design thinking with traditional engineering; articulating a personal starting point.	Workbook Weeks 1–3 (Activities 1–6, 12): starting-point sentence revised after the 5 Whys; DT vs traditional SE; five stages with CSE framing; Sheet M1.1.	**CT, YD**	CT-LO1, CT-LO6; IPD-LO3
CO2	Structured small-group discussion, observation and peer feedback; social constructivism.	Observation, empathy mapping, interviewing, survey design, synthesis into POV.	Activities 7–11, 13–14; Sheets M1.2, M2.1–M2.3; Sheets 1–2 (Problem Spotter, Empathy Map and Persona).	**RE, RWE, CT**	CT-LO2, CT-LO5; RE-LO2, RE-LO4; RWE-LO2, RWE-LO7
CO3	Problem-based learning with social learning theory; individual and collective ideation.	Divergent and convergent thinking; SCAMPER, mind-mapping, crazy-8s; framing HMW for a Transformer sub-component.	Sheets 3–5: HMW, Idea Generation (at least 8 ideas), Concept Sketch; social experience and market perspective capture (Ex III.1–III.2).	**IPD, CT**	CT-LO1, CT-LO6; IPD-LO3, IPD-LO4
CO4	Project-based prototyping; lo-fi to working proof of concept; mentor check-ins.	Dataset preparation, tokenisation, baseline and redesigned SLM training, version control, reproducible notebooks.	Sheet 6 (Lo-Fi V1), baseline SLM (Milestone M2), redesigned SLM and SLM v2 (Milestones M3–M4), Sheet 8 (Lo-Fi V2).	**IPD, RE**	IPD-LO1; RE-LO3
CO5	Peer review sessions, case study analysis, usability and A/B testing concepts.	Test planning, ablation, metrics for success, feedback integration, feasibility and SMART metrics.	Sheet 7 (Test Plan and Feedback Record), ablation study, feasibility and implementation plan (Ex IV.1–IV.3).	**CT, RE, RWE**	CT-LO3, CT-LO4; RE-LO3; RWE-LO9
CO6	Business model canvas workshops, pitch practice and transformative learning.	Business modelling, value proposition design, storytelling, responsible AI, pathway planning.	Sheets 9–11: Business Model Canvas, Pitch Storyboard, Final Pitch; model card; Track Pathway Declaration.	**IPD, RWE, YD**	IPD-LO3, IPD-LO4; RWE-LO8; YD-LO6

Abbreviations: CT – Critical Thinking; IPD – Innovation and Professional Development; RE – Research Experience; RWE – Real World Exposure; YD – Youth Development.

Session Wise Plan

The 60 contact hours are planned as 15 weeks of two 2-hour sessions (Theory/Studio and Lab). Weeks 1–3 use the Phase 1–2 workbook (Activities 1–14); Weeks 4–6 use Modules 1–2 sheets (M1.1–M2.3 and Handoff); Weeks 7–15 use Sheets 1–11 of the Modules 3–5 journal. PD – Planned Date; AD – Actual Date.

Wk	Ses	PD	AD	Unit	Topic and Activity (workbook / sheet reference)	Pedagogy	Real-World Exposure / SLM Milestone	Deliverable	CO
1	1			I	Introduction to Design Thinking: definition, principles, history, human-centred design fundamentals. Activities 1–5: My Starting Point (Page 1); Desirability–Feasibility–Viability; DT vs traditional software engineering; five stages with CSE framing; biomimicry detour.	Interactive lecture; Think-Pair-Share; experiential learning (Kolb cycle)	Technically correct applications that fail their users (CSE design lens: digital products and human-software interaction)	Workbook Page 1 and notes (A1–A5)	CO1
1	2			I	Activity 6: IDEO Shopping Cart deep dive (Page 2). Activity 7: Problem Observation Sheet (Page 3) on CSE Case Pack A–D (senior citizen banking, health-app logging, shopkeeper UPI, first-generation smartphone user).	Video analysis; case-based team work (individual first, then team consensus)	Observing real users before ideating; named people, workarounds, false design assumptions	Pages 2–3	CO1, CO2
2	3			I	Activity 8: Screenshot Analysis (Page 4) of government portals, banking and health interfaces. Activity 9: Five DT mindsets with CSE examples and team consensus (Page 5).	Structured observation; peer discussion	Real interface screenshots; who the design was NOT made for	Pages 4–5	CO1, CO2
2	4			I	Activity 10: Problem Spotter Sheet 1 – video/verbal scenario (Page 6). Activity 11: Problem Spotter Sheet 2 – text case (Page 7).	Individual-then-team observation; gallery share	Kiosk and patient-portal scenarios; WHO/WHAT/WHEN/WORKAROUND/CLUE	Pages 6–7	CO2
3	5			I	Activity 12: 5 Whys chain from symptom to root cause (Page 8); root cause as a false design assumption; revised starting-point sentence (Page 1).	Guided questioning; reflective observation	Root-cause thinking on digital-product failures	Page 8; revised Page 1	CO1, CO2
3	6			I	Activity 13: Problem Selection Matrix (Page 9). Activity 14: problem statement – first draft, peer feedback and final (Page 10).	Peer feedback practice; criteria-based selection	Scoring problems on human presence, observability, significance, designability, researchability	Pages 9–10	CO1, CO2
4	7			I	Module 1 (AIML lens). Sheet M1.1: Design Thinking Awareness Challenge – redesign the classroom experience; biomimicry spark; stages used. Design-thinking tools and methodologies; why generic AI models fail real users.	Think-Pair-Share; rapid design challenge	Classroom as a first user-research site	Sheet M1.1	CO1
4	8			I	Sheet M1.2: scout 2–3 candidate domains/datasets; mini empathy map. Sheet M1.3: draft HMW per domain; shortlisting rubric (curiosity, feasibility, impact /15). Primer: what a small language model (SLM) is; public dataset sources. Module 1 gate check.	Exploratory scouting; criteria-based decision making	SLM Milestone M0: shortlist 1–2 candidate domains with dataset source and licence	Sheets M1.2, M1.3; shortlist	CO1, CO2
5	9			II	Unit II – affective design and human-centred design process. Sheet M2.1: Emotional Journey Map for one end-to-end task (emotion, intensity 1–5, why).	Small-group discussion; social constructivism	Emotion mapping of real users working with the chosen data	Sheet M2.1	CO2
5	10			II	Sheet M2.2: structured interview design (question bank), interview notes log, detailed empathy map (feelings, thoughts, motivations). Interviews with at least one real or representative user between sessions.	Structured interviewing; peer review of interview guides	Practitioners/community members as interviewees; consent and anonymity	Sheet M2.2	CO2

Wk	Ses	PD	AD	Unit	Topic and Activity (workbook / sheet reference)	Pedagogy	Real-World Exposure / SLM Milestone	Deliverable	CO
6	11			II	Sheet M2.3: qualitative vs quantitative methods; 5-question mini survey; qualitative synthesis of recurring themes; user identification and segmentation.	Focused group discussion; synthesis workshop	Mini survey with real respondents	Sheet M2.3 (survey, themes)	CO2
6	12			II	Point-of-View statement; stakeholder map; Handoff Design Brief (dataset and source, domain focus, persona, POV, committed HMW, early hypothesis of where a vanilla Transformer struggles). Module 2 gate.	Peer feedback; synthesis	SLM Milestone M0 closes: dataset chosen; brief copied into Sheet 1	Handoff Design Brief; Unit II portfolio due	CO2, CO3
7	13			III	Module 3 launch. Transformer primer: embeddings, positional encoding, attention, feed-forward, normalisation, training objective as design surfaces. Sheet 1: Problem Spotter Sheet. Dataset audit.	Problem-based learning; concept walkthrough	SLM Milestone M1: dataset locked; data card (source, licence, size, consent and privacy checks)	Sheet 1; data card	CO2, CO4
7	14			III	Sheet 2: Empathy Map and User Persona. Sheet 3: How Might We – POV, three candidate HMWs, checklist; commit to one HMW aimed at a Transformer sub-component.	Peer critique of HMW; persona building	Persona grounded in Module 2 research, not assumed	Sheets 2–3	CO2, CO3
8	15			III	Unit III – ideation as a thinking process; individual vs collective ideation. Ex III.1: second-round individual interviews and journey maps. Sheet 4: Idea Generation (SCAMPER, mind-mapping, crazy-8s; at least 8 ideas across Transformer components).	Problem-based learning; social learning theory	Ideas generated across attention, positional encoding, tokenisation, objective, depth/width	Sheet 4 (draft); interview notes	CO3
8	16			III	Ex III.2: social experience capture and market perspective – existing tools, community practices, competing approaches. Converge ideas. Data cleaning, tokenizer choice.	Landscape scan; convergent selection	SLM Milestone M2 begins: cleaned corpus and tokenizer	Sheet 4 (converged); landscape note	CO3
9	17			III	Ex III.3: Sheet 5 Concept Sketch with numbered callouts; lo-fi vs hi-fi prototyping; train/validation/test split and training pipeline set-up.	Sketching; rapid prototyping	Reproducible pipeline (repository, notebook)	Sheet 5; pipeline repo	CO3, CO4
9	18			III	Sheet 6: Lo-Fi Prototype V1 – block-by-block redesign and hypothesis paragraph. Baseline SLM training run (vanilla model; loss curve, sample generations).	Hands-on prototyping	SLM Milestone M2: baseline SLM trained on the chosen dataset	Sheet 6; baseline results; Unit III portfolio due	CO4
10	19			IV	Unit IV – proof-of-concept review and evaluation methodologies; usability testing, qualitative/quantitative evaluation, A/B testing, metrics for success. Sheet 7: Test Plan (user-defined success plus SLM metrics).	Case study analysis; peer review	Success defined in the user's terms, not benchmark scores alone	Sheet 7 (plan)	CO5
10	20			IV	Implement the redesigned component(s) from Version 1 in the SLM pipeline; sanity tests; experiment log.	Hands-on prototyping; mentor check-in	SLM Milestone M3 begins	Working redesign branch; log	CO4
11	21			IV	Ex IV.1: PoC review gallery walk – peer and mentor feedback recorded on Sheet 7.	Peer review sessions	Structured feedback from users/mentors	Sheet 7 (feedback)	CO5
11	22			IV	Ablation run: baseline vs redesigned model on the same data and budget; record results, failures and surprises.	Experimental evaluation	SLM Milestone M3: ablation evidence	Ablation table	CO5
12	23			IV	Ex IV.2: implementation feasibility – technical (data, compute, size, latency), economic and social; SMART quality metrics; implementation plan.	Case study analysis; presentation practice	Deployment constraints of the target users	Feasibility and implementation plan	CO5
12	24			IV	Ex IV.3: testing methodologies and production planning. Sheet 8: Lo-Fi Prototype V2 revised from feedback. SLM v2 training.	Iterative design; peer review	SLM Milestone M4: SLM v2 evaluated; size and latency noted	Sheet 8; SLM v2; Unit IV portfolio due	CO4, CO5

Wk	Ses	PD	AD	Unit	Topic and Activity (workbook / sheet reference)	Pedagogy	Real-World Exposure / SLM Milestone	Deliverable	CO
13	25			V	Unit V – from design thinking to entrepreneurship. Ex V.1: connect design outcomes to business opportunities; customer segments. Sheet 9: Business Model Canvas (part 1). Pathway options: Professional, Entrepreneur, Research.	Business model canvas workshop	Who would deploy or benefit from the SLM	Sheet 9 (draft)	CO6
13	26			V	Sheet 9 completed; value proposition design; impact streams; model card draft.	Workshop; peer critique	SLM Milestone M5 begins: model card	Sheet 9; model card draft	CO6
14	27			V	Ex V.2: startup development planning – idea to startup; incubators, accelerators, funding mechanisms. Sheet 10: Pitch Storyboard (6 panels). Pitch practice.	Pitch presentation practice	Start-up ecosystem; success factors of design-driven ventures	Sheet 10; pitch plan	CO6
14	28			V	Ex V.3: social innovation and entrepreneurship applications; ethical dimensions and responsible AI (bias, privacy, consent); demo packaging.	Transformative learning; critical reflection	SLM Milestone M5: model card, runnable demo, responsible-AI note	Impact note; demo	CO6, CO3
15	29			V	Dress rehearsal; integration of hypothesis write-up, final annotated diagram and one-page canvas; draft Track Pathway Declaration.	Peer feedback; rehearsal	Rehearsal with external-style questions	Final package (draft)	CO3–CO6
15	30			V	Sheet 11: Final Pitch – Design Showcase (5–7 min per team) with live SLM demo; submission of design journal and repository; Track Pathway Declaration.	Presentation and defence	SLM Milestone M6: showcase and pathway declaration	Sheet 11; all 11 sheets; Unit V portfolio and project evaluation	CO1–CO6

Unit allocation: Unit I – Weeks 1–4 (16 h); Unit II – Weeks 5–6 (8 h); Unit III – Weeks 7–9 (12 h); Unit IV – Weeks 10–12 (12 h); Unit V – Weeks 13–15 (12 h). Total 60 h. The syllabus does not give a week-wise split; this allocation is proposed.

Small Language Model (SLM) Build Track

Each student (or pair/team of three) carries one dataset from Week 4 to Week 15. The SLM is the proof of concept for CO4 and the evidence base for CO5. Design decisions always come first: the architectural or training change must be traceable to the user insight in Sheets 1–3.

Milestone	Weeks	Link in the design journey	Student deliverable	Acceptance check
M0	4–6	Modules 1–2: scouting and research	Shortlist of 1–2 candidate domains with dataset source and licence	Dataset is public or licensed, domain-specific and text (or convertible to text); user group identified
M1	7	Sheets 1–2	Dataset locked; data card (source, licence, size, anonymisation and consent checks, known limits)	No personal or sensitive data without consent; licence permits training and demonstration
M2	8–9	Sheets 4–6 (Ideate, Prototype V1)	Cleaned corpus, tokenizer, train/validation/test split, baseline SLM trained	Reproducible notebook or repository; loss curve and sample generations
M3	10–11	Sheet 7 (Test)	Redesigned component implemented; ablation of baseline vs redesign on the same data and compute budget	At least one change from Sheet 6 implemented; results recorded, including failures
M4	12	Sheet 8 (Prototype V2); Ex IV.2–IV.3	SLM v2 trained using feedback; dataset-specific and user-defined metrics; size and latency noted	V2 changes trace to Sheet 7 feedback; feasibility note completed
M5	13–14	Sheets 9–10; Ex V.3	Model card (data, intended use, limits, bias); runnable demo or notebook; responsible-AI note	A peer can run the demo from the repository instructions

Milestone	Weeks	Link in the design journey	Student deliverable	Acceptance check
M6	15	Sheet 11	Showcase with live or recorded demo; repository hand-over; Track Pathway Declaration	Claims in the pitch match the recorded evidence

Scale is set by the faculty after a compute check: for example a small transformer trained from scratch on the chosen corpus, or parameter-efficient fine-tuning of an open small model, run on institutional or free GPU resources. The grade depends on the quality of the design reasoning and evidence, not on model size or benchmark score.

Track Pathway Bridge

Track	What the student carries forward	Linked course / semester	Evidence at Week 15
Professional	Deployable portfolio artefact: trained SLM, repository, demo and model card	Track-based specialisations (syllabus)	Repository, demo, model card
Entrepreneur	Validated persona, Business Model Canvas, pitch and SLM as a minimum viable product	Product Innovation & Entrepreneurship (PCC 311)	Sheets 9–11, one-page canvas, pitch deck
Research	Hypothesis, ablation evidence and open questions for a reproducible study	Research Methods and Design (PCC 306)	Hypothesis write-up, ablation tables, reproducibility notes

Assessment Strategy Design

Overall Assessment Distribution

Assessment Component	Assessment Type	Weightage (%)	Purpose
Unit-wise Exercise Portfolio	Practical exercises and documentation (five units, three exercises per unit)	80	Builds and records the skills of each design-thinking stage through worksheets, sheets and the SLM build track
Comprehensive Design Challenge Project	Integrative project with presentation	20	Integrates the journey into a hypothesis write-up, annotated architecture, SLM evidence and a final pitch
Total	**100% Continuous Assessment**	**100**	**No semester-end examination**

Formative Assessment (Continuous)

Assessment Tool	Weightage	Frequency	COs Assessed	Individualization
Unit I Portfolio – DT Foundations (Weeks 1–4)	20	3 exercises	CO1, CO2	Individual worksheets; team consensus pages marked separately
Unit II Portfolio – Human-Centred and Affective Design (Weeks 5–6)	12	3 exercises	CO2, CO3	Individual interview log; team survey and brief
Unit III Portfolio – Ideation and Conceptualisation (Weeks 7–9)	16	3 exercises	CO2, CO3, CO4	Own dataset and persona; team of up to three with identified contributions

Assessment Tool	Weightage	Frequency	COs Assessed	Individualization
Unit IV Portfolio – Confirmation and Review (Weeks 10–12)	16	3 exercises	CO4, CO5	Individual ablation log; peer review as a team
Unit V Portfolio – Entrepreneurship and Social Innovation (Weeks 13–15)	16	3 exercises	CO3, CO6	Individual model card and canvas section; team pitch
Comprehensive Design Challenge Project (Week 15)	20	1	CO1–CO6	Team pitch plus individual contribution statement and questions
Total	**100**			

A. Unit I Portfolio – Design Thinking Foundations (20 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Ex I.1 Design Thinking Awareness: starting-point sentence and its revision, DT vs traditional SE comparison, five stages with CSE framing, biomimicry example, IDEO observation notes, classroom-redesign challenge	Activities 1–6 (Pages 1–2); Sheet M1.1	CT-LO1, CT-LO6 (primary); IPD-LO3 (supports)	CO1	6
Ex I.2 Stage 1 – Empathise: problem observation sheet from the CSE Case Pack, screenshot analysis, five-mindset consensus, Problem Spotter Sheets 1–2, scouting of candidate domains with a mini empathy map	Activities 7–11 (Pages 3–7); Sheet M1.2	CT-LO2, CT-LO5 (primary); RWE-LO2, RWE-LO3, RWE-LO7 (supports)	CO2 (primary); CO1	7
Ex I.3 Stage 2 – Define: 5 Whys to a root cause, problem selection matrix, first-draft and peer-reviewed final problem statement, draft HMW per domain and shortlisting rubric	Activities 12–14 (Pages 8–10); Sheet M1.3	CT-LO1, CT-LO4 (primary); IPD-LO3 (supports)	CO1, CO2	7
Subtotal				**20**

B. Unit II Portfolio – Human-Centred and Affective Design (12 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Ex II.1 Observing emotional aspects: emotion mapping and emotional journey map of one real end-to-end task	Sheet M2.1	CT-LO5, RWE-LO7 (primary)	CO2	4
Ex II.2 Empathy and emotional behaviour analysis: structured interview with at least one real or representative user, interview notes log, detailed empathy map (feelings, thoughts, motivations)	Sheet M2.2	RE-LO4, RWE-LO1, YD-LO2 (primary)	CO2	4
Ex II.3 User research and POV: five-question mini survey, qualitative synthesis, point-of-view statement, stakeholder map and Design Brief hand-off	Sheet M2.3; Handoff Design Brief	RE-LO2, RE-LO3, CT-LO4 (primary); IPD-LO3 (supports)	CO2, CO3	4
Subtotal				**12**

C. Unit III Portfolio – Ideation and Conceptualisation (16 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Ex III.1 Individual user interviews and perspective gathering: Problem Spotter Sheet (dataset, domain, at least three gaps, where a vanilla Transformer struggles), persona and empathy map, data card with source, licence, consent and privacy checks	Sheets 1–2; SLM Milestone M1	CT-LO2, RE-LO2, RWE-LO8 (primary)	CO2, CO3	5
Ex III.2 Social experience capture and market perspective: committed HMW (POV, three candidates, checklist) and landscape scan of existing tools and community practice	Sheet 3; landscape note	CT-LO3, RWE-LO2, YD-LO6 (primary)	CO3	5
Ex III.3 Systematic ideation and proof of concept: at least eight ideas across Transformer components, annotated concept sketch, Lo-Fi V1 with hypothesis; cleaned dataset, tokenizer and baseline SLM run	Sheets 4–6; SLM Milestone M2	IPD-LO1, IPD-LO3, IPD-LO4, CT-LO6 (primary)	CO3, CO4	6
Subtotal				**16**

D. Unit IV Portfolio – Confirmation and Review (16 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Ex IV.1 Proof-of-concept review and evaluation: test plan with user-defined and SLM metrics; structured peer and mentor review of Version 1 and baseline results	Sheet 7	CT-LO4, RWE-LO3 (primary)	CO5	5
Ex IV.2 Implementation feasibility assessment: technical (data, compute, size, latency), economic and social feasibility; SMART quality metrics; implementation plan	Feasibility and implementation plan (added annex to Sheet 8)	CT-LO3, RWE-LO9, IPD-LO1 (primary)	CO5	5
Ex IV.3 Testing methodologies and production planning: ablation (baseline vs redesign), failure analysis, Lo-Fi V2 revised from feedback, SLM v2, basic deployment feasibility	Sheet 8; SLM Milestones M3–M4	RE-LO3, CT-LO1, IPD-LO1 (primary)	CO4, CO5	6
Subtotal				**16**

E. Unit V Portfolio – Design Thinking to Entrepreneurship (16 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Ex V.1 Connecting design outcomes to business opportunities: Business Model Canvas, target segments, value proposition and track mapping	Sheet 9	IPD-LO3, RWE-LO2, RWE-LO9 (primary)	CO6	5
Ex V.2 Start-up development planning: pitch storyboard, pitch deck or plan, note on incubators, accelerators and funding	Sheet 10	IPD-LO4, RWE-LO3, RWE-LO6 (primary)	CO6	5
Ex V.3 Social innovation and entrepreneurship applications: social-impact and responsible-AI note (bias, privacy, consent), model card, demo, Track Pathway Declaration	Model card; impact note; SLM Milestone M5	YD-LO6, RWE-LO8, CT-LO4 (primary)	CO6, CO3	6
Subtotal				**16**

F. Comprehensive Design Challenge Project (20 marks)

Exercise (what is assessed and the evidence)	Worksheet / sheet basis	LO mapping	CO	Marks
Depth of empathy and problem framing: real user insight, not an assumed persona; HMW traceable to that insight	Sheets 1–3; showcase rubric (25%)	CT-LO2, RE-LO2 (primary)	CO2	5
Quality and justification of the redesign: each change tied to a specific limitation of a vanilla Transformer for the dataset; baseline vs redesigned SLM evidence	Sheets 5–6, 8; showcase rubric (30%)	CT-LO1, IPD-LO3 (primary)	CO3, CO4	6
Rigour of test plan and iteration: metrics fit the user's definition of success; ablation; Version 2 visibly responds to feedback	Sheets 7–8; showcase rubric (20%)	RE-LO3, CT-LO4 (primary)	CO5	4
Clarity of hypothesis write-up and diagram: a reader unfamiliar with the dataset can follow the argument from problem to architecture	Final write-up and annotated diagram; showcase rubric (15%)	CT-LO4 (primary)	CO3, CO4	3
Pitch and business framing: how the redesign aids SLM training, who benefits, and the chosen track	Sheets 9–11; showcase rubric (10%)	IPD-LO4, RWE-LO3 (primary)	CO6	2
Subtotal				**20**

The Phase 4 showcase rubric weights (25/30/20/15/10) are applied to the 20 project marks. Unit portfolios assess each exercise worksheet and the process evidence; the project assesses the integrated journey and final pitch. The faculty should confirm that evidence shared between the two components is not counted twice.

Assessment Calendar

Sl. No.	Assessment Week	Assessment Instrument	Marks	Description	COs Covered
1	Week 4 (Sessions 1–8)	Unit I Portfolio	20	Activities 1–14 and Sheets M1.1–M1.3; Module 1 gate: shortlist of 1–2 domains.	CO1, CO2
2	Week 6 (Sessions 9–12)	Unit II Portfolio	12	Sheets M2.1–M2.3 and Handoff Design Brief; Module 2 gate: dataset chosen.	CO2, CO3
3	Week 9 (Sessions 13–18)	Unit III Portfolio	16	Sheets 1–6, data card, baseline SLM (Milestones M1–M2).	CO2, CO3, CO4
4	Week 12 (Sessions 19–24); peer review in Week 11	Unit IV Portfolio	16	Sheets 7–8, feasibility plan, ablation, SLM v2 (Milestones M3–M4).	CO4, CO5
5	Week 15 (Sessions 25–30)	Unit V Portfolio	16	Sheets 9–10, model card, impact note, Track Pathway Declaration (Milestone M5).	CO3, CO6
6	Week 15 (Session 30)	Comprehensive Design Challenge Project	20	Final pitch (5–7 min per team) with SLM demo; submission of all 11 sheets, write-up, diagram and repository (Milestone M6).	CO1–CO6
		Total	**100**		**CO1–CO6**

Course Outcomes (CO) to Components Mapping

CO	Unit I	Unit II	Unit III	Unit IV	Unit V	Project	Primary evidence
CO1	✓✓	✓				✓	Starting-point sentence and revision; DT vs SE; five-stage notes
CO2	✓✓	✓✓	✓			✓✓	Observation sheets, interviews, journey map, persona, survey
CO3		✓	✓✓		✓	✓✓	HMW, idea sheet, concept sketch, social innovation note
CO4			✓✓	✓✓		✓✓	Lo-Fi V1 and V2, baseline SLM, redesigned SLM
CO5			✓	✓✓		✓✓	Test plan, ablation, feedback record, feasibility plan
CO6				✓	✓✓	✓✓	Business Model Canvas, pitch, model card, pathway declaration

Legend: ✓✓ = Primary/direct assessment | ✓ = Supporting/reinforcing

Dimension Focus by Unit

Unit	Primary dimensions	Supporting dimensions	Basis
Unit I	CT, YD	RWE, IPD	Sheet M1.1 (CT, YD), M1.2 (CT, RWE), M1.3 (CT, IPD)
Unit II	RE, RWE	CT, YD	Sheets M2.1 (RWE, YD), M2.2 (RE, RWE), M2.3 (RE, CT, IPD)
Unit III	IPD, CT	RE, RWE	Ideation, concept sketch and prototype; dataset ethics and market perspective
Unit IV	CT, RE	IPD, RWE	Test plan, ablation, feasibility and production planning
Unit V	IPD, RWE	YD, CT	Business model, pitch, social innovation and responsible AI
Project	CT, IPD	RE, RWE	Integrated journey and final pitch

Dimensions follow the tags of the Phase 3 sheets. The syllabus and the supplied material do not give a BoS-approved percentage split for this course, so none is stated here.

Year-Level Learning Outcomes Coverage

Learning Outcome	Unit I	Unit II	Unit III	Unit IV	Unit V	Project
CT-LO1 – Challenge existing knowledge through structured analysis	✓		✓	✓		✓
CT-LO2 – Apply systematic observation methods	✓	✓		✓		✓
CT-LO3 – Evaluate technical information from multiple sources			✓	✓		✓
CT-LO4 – Develop reasoned arguments and corroborate using AI	✓	✓	✓	✓	✓	✓
CT-LO5 – Perceive objects and knowledge	✓	✓				
CT-LO6 – Raise doubts on status quo	✓		✓			✓
IPD-LO1 – Execute project-based learning assignments independently			✓	✓	✓	✓

Learning Outcome	Unit I	Unit II	Unit III	Unit IV	Unit V	Project
IPD-LO3 – Understand design thinking and its cultural and social implications	✓	✓	✓		✓	✓
IPD-LO4 – Use Generative AI to visualize real-world problems			✓		✓	✓
RE-LO2 – Define, identify, justify and articulate research problems		✓	✓			✓
RE-LO3 – Structure mini-projects using appropriate research methods		✓		✓		✓
RE-LO4 – Communicate with industry experts and researchers		✓				
RWE-LO1 – Interpersonal and social skills with stakeholders		✓				
RWE-LO2 – Observe real-world society, industry and market	✓	✓	✓		✓	
RWE-LO3 – Group discussion and social communication	✓			✓	✓	✓
RWE-LO7 – Affective and cultural aspects of design thinking	✓	✓				
RWE-LO8 – Professional standards and ethics			✓		✓	✓
RWE-LO9 – Awareness of industrial practices				✓	✓	
YD-LO2 – Interact with the public, community leaders and NGOs		✓				
YD-LO6 – Awareness of social implications of engineering solutions using AI			✓		✓	✓

Only learning outcomes with direct evidence in the units are marked. Learning outcomes not listed are not directly assessed in this course.

AI Integration Strategy

Level 1 – PROHIBITED: No AI tools; requires authentic personal work (observation, reflection, interviews).

Level 2 – PERMITTED (with disclosure): AI allowed for specific purposes and disclosed in the submission.

Level 3 – ENCOURAGED (with disclosure): AI promoted as a learning and building tool; use is documented and verified.

Component	Policy Level	Rationale	Disclosure Required
Unit I Portfolio	Level 1: Prohibited	The course begins with what the student personally notices; the starting-point sentence, observations and 5 Whys must be the student's own thinking.	NA
Unit II Portfolio	Level 1 for field work; Level 2 for drafting	Interviews, emotion mapping and field notes must come from real people. AI may help draft survey questions or tidy transcripts.	Tool, purpose and what was changed after checking; interview content never AI-generated
Unit III Portfolio	Level 3: Encouraged	AI can widen ideation, scan the landscape and scaffold pipeline code, but the persona and empathy content must come from real research.	AI record: tool, prompt purpose, suggestion, verification, final decision
Unit IV Portfolio	Level 3 for code and analysis; Level 1 for feedback	AI may support code, plots and analysis. Peer and mentor feedback must be collected from humans.	AI record; ablation code reviewed and results reproduced by the student
Unit V Portfolio	Level 2: Permitted	AI may help with canvas drafts and pitch polishing; the impact and ethics reflection is personal.	Disclosure statement in the submission
Comprehensive Design Challenge Project	Level 3: Encouraged	AI is part of the SLM workflow. Students remain responsible for every design decision and line of code and are questioned on both at the showcase.	AI use statement in the design journal and repository; unverified AI claims earn little credit

Course Facilitator